ITA0608 - Machine Learning
List of Lab Exercises

3.Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and applying this knowledge to classify a new sample.

Aim

To implement and demonstrate the ID3 Decision Tree Algorithm using a training dataset, construct a decision tree based on information gain, and classify a new sample.

Algorithm

1. Import the required libraries.
2. Load the Play Tennis dataset from a CSV file.
3. Preprocess the dataset if necessary.
4. Train the Decision Tree model using the ID3 algorithm.
5. Construct the decision tree based on the attribute with the highest information gain.
6. Input a new sample for testing.
7. Predict the class label of the new sample.
8. Display the prediction result.

Program Code

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.tree import DecisionTreeClassifier, plot_tree

# Read dataset directly as 'play_tennis_dataset.xlsx' is available

data = pd.read_excel("play_tennis_dataset.xlsx")

print("\nOriginal Dataset\n")

print(data)

# Convert categorical values into numeric values

encoded_data = data.copy()
```

```python
encoding_dictionary = {}

for column in encoded_data.columns:

    unique_values = encoded_data[column].unique()

    value_map = {value:index for index,value in enumerate(unique_values)}

    encoding_dictionary[column] = value_map

    encoded_data[column] = encoded_data[column].map(value_map)

print("\nEncoded Dataset\n")

print(encoded_data)

# Separate features and target

features = encoded_data.drop("PlayTennis", axis=1)

target = encoded_data["PlayTennis"]

# Build Decision Tree using Entropy (ID3)

classifier = DecisionTreeClassifier(

    criterion="entropy",

    random_state=10

)

classifier.fit(features, target)

print("\nModel Training Completed Successfully.")

# Draw Decision Tree

plt.figure(figsize=(13,8))

plot_tree(

    classifier,

    feature_names=features.columns,

    class_names=["No","Yes"],

    filled=True,
```

```python
    rounded=True,
    fontsize=10
)
plt.title("ID3 Decision Tree")
plt.show()
sample = {
    "Outlook":"Sunny",
    "Temperature":"Cool",
    "Humidity":"High",
    "Wind":"Weak"
}
sample_df = pd.DataFrame([sample])
for column in sample_df.columns:
    sample_df[column] = sample_df[column].map(encoding_dictionary[column])
prediction = classifier.predict(sample_df)
reverse_output = {v:k for k,v in encoding_dictionary["PlayTennis"].items()}
print("\n========== Prediction ==========")
print("Input Sample")
print(sample)
print("--------------------------------")
print("Predicted Class :", reverse_output[prediction[0]])
print("================================")
```

Output

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier, plot_tree

# Read dataset directly as 'play_tennis_dataset.xlsx' is available
data = pd.read_excel("play_tennis_dataset.xlsx")

print("\nOriginal Dataset\n")
print(data)

# Convert categorical values into numeric values
encoded_data = data.copy()

encoding_dictionary = {}

for column in encoded_data.columns:
    unique_values = encoded_data[column].unique()
    value_map = {value:index for index,value in enumerate(unique_values)}
    encoding_dictionary[column] = value_map
    encoded_data[column] = encoded_data[column].map(value_map)

print("\nEncoded Dataset\n")
print(encoded_data)

# Separate features and target
features = encoded_data.drop("PlayTennis", axis=1)
target = encoded_data["PlayTennis"]

# Build Decision Tree using Entropy (ID3)
classifier = DecisionTreeClassifier(
    criterion="entropy",
    random_state=10
)

classifier.fit(features, target)

print("\nModel Training Completed Successfully.")
```

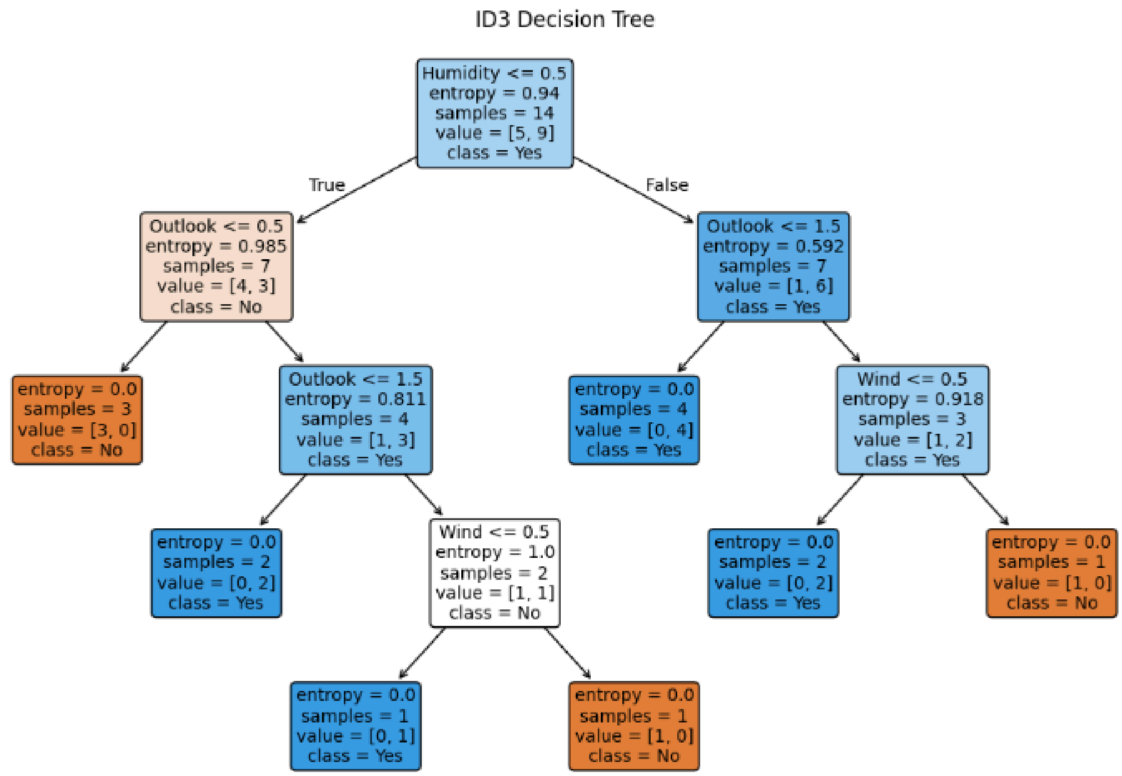
```python
plt.figure(figsize=(13,8))

plot_tree(
    classifier,
    feature_names=features.columns,
    class_names=["No","Yes"],
    filled=True,
    rounded=True,
    fontsize=10
)

plt.title("ID3 Decision Tree")
plt.show()

# ------------------------------
# Test New Sample
# ------------------------------

sample = {
    "Outlook":"Sunny",
    "Temperature":"Cool",
    "Humidity":"High",
    "Wind":"Weak"
}

sample_df = pd.DataFrame([sample])

for column in sample_df.columns:
    sample_df[column] = sample_df[column].map(encoding_dictionary[column])

prediction = classifier.predict(sample_df)

reverse_output = {v:k for k,v in encoding_dictionary["PlayTennis"].items()}

print("\n========== Prediction ==========")
print("Input Sample")
print(sample)
print("-------------------------------")
print("Predicted Class :", reverse_output[prediction[0]])
print("===============================")
```

```python
print("==============================")
```

```
Original Dataset

     Outlook Temperature Humidity    Wind PlayTennis
0      Sunny         Hot     High    Weak         No
1      Sunny         Hot     High  Strong         No
2   Overcast         Hot     High    Weak        Yes
3       Rain        Mild     High    Weak        Yes
4       Rain        Cool   Normal    Weak        Yes
5       Rain        Cool   Normal  Strong         No
6   Overcast        Cool   Normal  Strong        Yes
7      Sunny        Mild     High    Weak         No
8      Sunny        Cool   Normal    Weak        Yes
9       Rain        Mild   Normal    Weak        Yes
10     Sunny        Mild   Normal  Strong        Yes
11  Overcast        Mild     High  Strong        Yes
12  Overcast         Hot   Normal    Weak        Yes
13      Rain        Mild     High  Strong         No

Encoded Dataset

    Outlook  Temperature  Humidity  Wind  PlayTennis
0         0            0         0     0           0
1         0            0         0     1           0
2         1            0         0     0           1
3         2            1         0     0           1
4         2            2         1     0           1
5         2            2         1     1           0
6         1            2         1     1           1
7         0            1         0     0           0
8         0            2         1     0           1
9         2            1         1     0           1
10        0            1         1     1           1
11        1            1         0     1           1
12        1            0         1     0           1
13        2            1         0     1           0

Model Training Completed Successfully.
```

ID3 Decision Tree

```
13        2            1          0     1           0

Model Training Completed Successfully.
```




```
========== Prediction ==========
Input Sample
{'Outlook': 'Sunny', 'Temperature': 'Cool', 'Humidity': 'High', 'Wind': 'Weak'}
------------------------------
Predicted Class : No
==============================
```

Result

The ID3 Decision Tree Algorithm was successfully implemented using the Play Tennis dataset. A decision tree was generated based on information gain, and the new sample:

Outlook = Sunny, Temperature = Cool, Humidity = High, Wind = Weak

was successfully classified as:

Prediction: No (Do Not Play Tennis)

Thus, the ID3 algorithm correctly built the decision tree and classified the new sample based on the learned rules.